

Vectors and Linear Algebra

Mathematics for Computing

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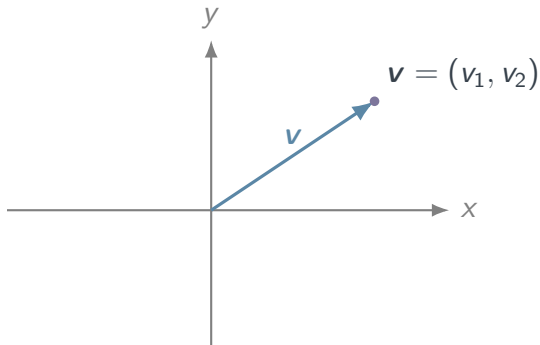
- ▶ Determinants
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Vectors

Objects used to represent magnitude, direction, position, and features



What Is a Vector?

A **vector** is a mathematical object that has:

- ▶ magnitude
- ▶ direction

A vector in two-dimensional space can be represented as

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}.$$

Key Idea

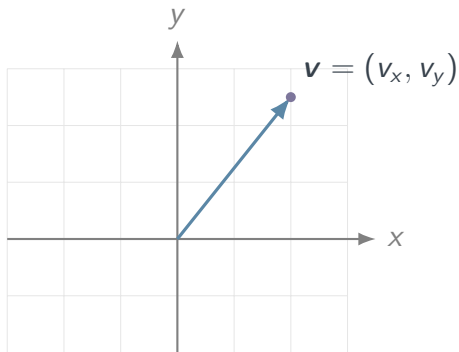
The entries of a vector are its **components** with respect to a chosen coordinate system.

Coordinate Systems

A coordinate system provides a reference for describing vectors and points.

For the standard two-dimensional Cartesian system:

$$\mathbf{v} = \begin{bmatrix} v_x \\ v_y \end{bmatrix}$$



Representing Vectors

Vectors are commonly written as column vectors:

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} \in \mathbb{R}^3.$$

More generally,

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} \in \mathbb{R}^n.$$

Interpretation

A vector in $\mathbb{R}^n$ has n components and can be viewed as a point or a directed quantity in an n -dimensional space.

Vector Addition

Two vectors of the same dimension can be added component-wise.

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix}$$

give

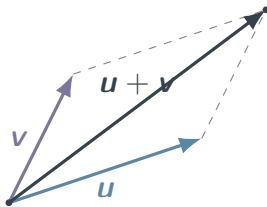
$$\mathbf{u} + \mathbf{v} = \begin{bmatrix} u_1 + v_1 \\ u_2 + v_2 \end{bmatrix}$$

Example

$$\begin{bmatrix} 2 \\ 1 \end{bmatrix} + \begin{bmatrix} 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}$$

Geometric Interpretation of Addition

Vector addition can be visualized using the **parallelogram rule**.



Scalar Multiplication

A **scalar** is a number that can scale a vector.

For a scalar c ,

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$$

$$c\mathbf{v} = \begin{bmatrix} cv_1 \\ cv_2 \\ \vdots \\ cv_n \end{bmatrix}$$

- ▶ $|c| > 1$: vector is stretched
- ▶ $0 < |c| < 1$: vector is shortened
- ▶ $c < 0$: direction is reversed

Magnitude of a Vector

The magnitude or Euclidean norm measures the length of a vector.

For

$$\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \end{bmatrix} \quad \boxed{\|\mathbf{v}\| = \sqrt{v_1^2 + v_2^2}}$$

For $\mathbf{v} \in \mathbb{R}^n$,

$$\boxed{\|\mathbf{v}\| = \sqrt{\sum_{i=1}^n v_i^2}}$$

Example

$$\mathbf{v} = \begin{bmatrix} 3 \\ 4 \end{bmatrix} \implies \|\mathbf{v}\| = \sqrt{3^2 + 4^2} = 5$$

Distance Between Vectors

The distance between two vectors is the magnitude of their difference.

For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$,

$$d(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$$

For example,

$$\mathbf{u} = \begin{bmatrix} 3 \\ 4 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

give

$$d(\mathbf{u}, \mathbf{v}) = \left\| \begin{bmatrix} 2 \\ 3 \end{bmatrix} \right\| = \sqrt{13}.$$

Vectors in Computing

Vectors provide a common representation for numerical information.

Machine Learning

Feature vectors represent measurable properties of an observation.

Computer Graphics

Coordinates and transformations can be represented using vectors.

Data Representation

Objects such as documents or records can be represented numerically.

Real-world information $\longrightarrow$ Numerical vector

Dot Product

For two vectors

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix},$$

their **dot product** is

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i$$

or equivalently,

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$$

Dot Product: Example

Consider

$$\mathbf{u} = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 4 \\ -1 \end{bmatrix}.$$

Then

$$\begin{aligned} \mathbf{u} \cdot \mathbf{v} &= (2)(4) + (3)(-1) \\ &= 8 - 3 \\ &= 5. \end{aligned}$$

Therefore,

$$\boxed{\mathbf{u} \cdot \mathbf{v} = 5}$$

The dot product of two vectors is a **scalar**, not a vector.

Geometric Interpretation of the Dot Product

The dot product can also be expressed as

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

where θ is the angle between the vectors.

Therefore,

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

The dot product therefore connects:

algebra $\longleftrightarrow$ geometry.

Angle Between Two Vectors

For non-zero vectors $\mathbf{u}$ and $\mathbf{v}$,

$$\theta = \cos^{-1} \left(\frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|} \right)$$

The sign of the dot product provides useful geometric information:

- ▶ $\mathbf{u} \cdot \mathbf{v} > 0$: acute angle
- ▶ $\mathbf{u} \cdot \mathbf{v} = 0$: right angle
- ▶ $\mathbf{u} \cdot \mathbf{v} < 0$: obtuse angle

Orthogonal Vectors

Two vectors are **orthogonal** if they are perpendicular.

For vectors $\mathbf{u}$ and $\mathbf{v}$,

$$\mathbf{u} \cdot \mathbf{v} = 0$$

implies

$$\mathbf{u} \perp \mathbf{v}$$

provided both vectors are non-zero.

Example

$$\mathbf{u} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ -2 \end{bmatrix}$$

Since

$$\mathbf{u} \cdot \mathbf{v} = 2(1) + 1(-2) = 0,$$

the vectors are orthogonal.

Orthogonal Projection

The projection of $\mathbf{u}$ onto a non-zero vector $\mathbf{v}$ is

$$\text{proj}_{\mathbf{v}}(\mathbf{u}) = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}$$

The projection gives the component of $\mathbf{u}$ that lies in the direction of $\mathbf{v}$.

It is useful in:

- ▶ geometry
- ▶ least-squares methods
- ▶ computer graphics
- ▶ machine learning

Decomposing a Vector

A vector can be separated into components parallel and perpendicular to another vector.

$$\mathbf{u} = \text{proj}_{\mathbf{v}}(\mathbf{u}) + \mathbf{u}_{\perp}$$

where

$$\mathbf{u}_{\perp} = \mathbf{u} - \text{proj}_{\mathbf{v}}(\mathbf{u}).$$

The two components satisfy

$$\mathbf{u}_{\perp} \cdot \mathbf{v} = 0.$$

Dot Product and Distance

The dot product can also be used to express the squared distance between two vectors.
For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$,

$$\|\mathbf{u} - \mathbf{v}\|^2 = (\mathbf{u} - \mathbf{v})^T (\mathbf{u} - \mathbf{v}).$$

Expanding,

$$\|\mathbf{u} - \mathbf{v}\|^2 = \mathbf{u}^T \mathbf{u} + \mathbf{v}^T \mathbf{v} - 2\mathbf{u}^T \mathbf{v}$$

Thus the dot product is closely connected to both:

length and distance.

Dot Product and Duality

For a fixed vector $\mathbf{w}$, consider

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x}.$$

This takes a vector $\mathbf{x}$ and produces a scalar.

$$\boxed{\mathbf{x} \longrightarrow \mathbf{w}^T \mathbf{x}}$$

Such a function is a **linear functional**.

This provides a connection between:

$$\boxed{\text{vectors} \longleftrightarrow \text{linear functions}}$$

and leads naturally to the idea of **duality**.

Weight Vectors

A weight vector can define a linear function

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x}.$$

For

$$\mathbf{w} = \begin{bmatrix} w_1 \\ w_2 \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

we obtain

$$f(\mathbf{x}) = w_1 x_1 + w_2 x_2.$$

In machine learning, weight vectors are fundamental to:

- ▶ linear models
- ▶ linear classifiers
- ▶ perceptrons

Hyperplanes

A vector $\mathbf{w}$ can define a hyperplane through

$$\mathbf{w}^T \mathbf{x} = c$$

where c is a constant.

The vector $\mathbf{w}$ is **normal** to the hyperplane.

In two dimensions:

$$w_1 x_1 + w_2 x_2 = c$$

represents a line.

In three dimensions:

$$w_1 x_1 + w_2 x_2 + w_3 x_3 = c$$

represents a plane.

Hyperplanes in Classification

A linear classifier can use

$$f(\mathbf{x}) = \mathbf{w}^T \mathbf{x} + b.$$

The decision boundary is

$$\boxed{\mathbf{w}^T \mathbf{x} + b = 0}$$

The space can then be divided according to the sign of $f(\mathbf{x})$:

$$\begin{cases} f(\mathbf{x}) > 0 & \text{one side} \\ f(\mathbf{x}) < 0 & \text{other side.} \end{cases}$$

This is the geometric foundation of models such as the **perceptron**.

Linear Combinations

A **linear combination** of vectors is formed by multiplying vectors by scalars and adding the results.

For vectors $\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k$,

$$\mathbf{w} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k$$

where $c_1, c_2, \dots, c_k$ are scalars.

The scalars are called the **coefficients** of the linear combination.

Linear Combination: Example

Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Consider

$$2\mathbf{u} - \mathbf{v}.$$

Then

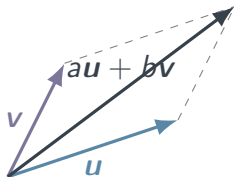
$$\begin{aligned} 2\mathbf{u} - \mathbf{v} &= 2 \begin{bmatrix} 1 \\ 2 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\ &= \begin{bmatrix} 2 \\ 4 \end{bmatrix} - \begin{bmatrix} 3 \\ 1 \end{bmatrix} \\ &= \boxed{\begin{bmatrix} -1 \\ 3 \end{bmatrix}} \end{aligned}$$

Linear Combinations: Geometric View

Suppose

$$w = au + bv.$$

The coefficients a and b determine how much of each vector contributes to w .



$$w = au + bv$$

Span of Vectors

The **span** of a set of vectors is the set of all possible linear combinations of those vectors.

For

$$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_k,$$

the span is

$$\text{span}\{\mathbf{v}_1, \dots, \mathbf{v}_k\} = \left\{ \sum_{i=1}^k c_i \mathbf{v}_i : c_i \in \mathbb{R} \right\}$$

The span describes the set of vectors that can be generated from the given vectors.

Span in $\mathbb{R}^2$

Consider two vectors

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Any vector

$$\begin{bmatrix} x \\ y \end{bmatrix} \in \mathbb{R}^2$$

can be written as

$$\begin{bmatrix} x \\ y \end{bmatrix} = x \begin{bmatrix} 1 \\ 0 \end{bmatrix} + y \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Therefore,

$$\boxed{\text{span}\{\mathbf{u}, \mathbf{v}\} = \mathbb{R}^2}$$

Geometric Meaning of Span

The span of vectors can have different dimensions.

One Direction

A non-zero vector spans a **line**.

$$\text{span}\{\mathbf{v}\}$$

Two Directions

Two independent vectors in $\mathbb{R}^2$ span a **plane**.

$$\text{span}\{\mathbf{u}, \mathbf{v}\}$$

Three Directions

Three independent vectors in $\mathbb{R}^3$ span three-dimensional space.

$$\mathbb{R}^3$$

The span describes the **space generated by the vectors**.

Linearly Independent Vectors

Vectors $\mathbf{v}_1, \dots, \mathbf{v}_k$ are **linearly independent** if the equation

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \dots + c_k \mathbf{v}_k = \mathbf{0}$$

has only the trivial solution

$$c_1 = c_2 = \dots = c_k = 0.$$

In other words, none of the vectors can be written as a linear combination of the others.

Linearly Dependent Vectors

Vectors are **linearly dependent** if there exists a non-trivial set of coefficients such that

$$c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2 + \cdots + c_k \mathbf{v}_k = \mathbf{0},$$

where at least one coefficient is non-zero.

Non-trivial solution $\implies$ Linear dependence

This means that at least one vector can be expressed as a linear combination of the others.

Linear Dependence: Example

Consider

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Notice that

$$\mathbf{v} = 2\mathbf{u}.$$

Therefore,

$$2\mathbf{u} - \mathbf{v} = \mathbf{0}.$$

The coefficients

$$2 \neq 0, \quad -1 \neq 0$$

give a non-trivial solution.

Hence,

$\mathbf{u}, \mathbf{v}$ are linearly dependent.

Linear Independence: Example

Consider

$$\mathbf{u} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Suppose

$$a\mathbf{u} + b\mathbf{v} = \mathbf{0}.$$

Then

$$a \begin{bmatrix} 1 \\ 0 \end{bmatrix} + b \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Therefore,

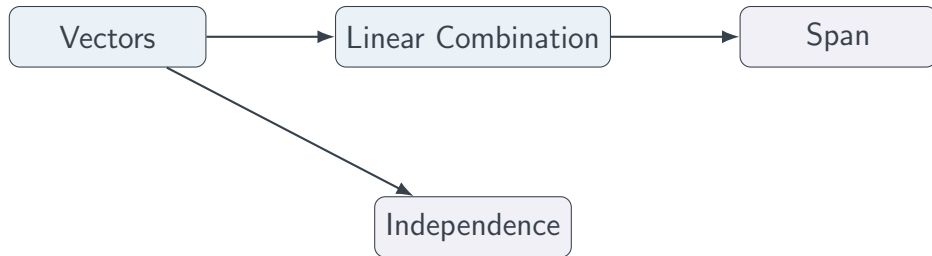
$$a = 0, \quad b = 0.$$

Hence,

$\mathbf{u}, \mathbf{v}$ are linearly independent.

How the Concepts Are Connected

The ideas introduced so far are closely related:



- ▶ Linear combinations generate vectors.
- ▶ The span contains all vectors that can be generated.
- ▶ Linear independence describes whether the generators are redundant.

Linear Combinations and Linear Systems

Consider

$$Ax = b.$$

If

$$A = \left[\begin{array}{c|c|c|c} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & & | \end{array} \right],$$

then

$$Ax = x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n.$$

Therefore,

$$\boxed{Ax = b}$$

asks whether $\mathbf{b}$ can be expressed as a linear combination of the columns of A .

This connection will become important when we study **column space** and **systems of linear equations**.

Independent Vectors and Dimension

In $\mathbb{R}^n$, there can be at most n linearly independent vectors.

For example:

$\mathbb{R}^2 \Rightarrow$ at most 2 independent vectors

$\mathbb{R}^3 \Rightarrow$ at most 3 independent vectors

$\mathbb{R}^n \Rightarrow$ at most n independent vectors.

This observation leads naturally to the concepts of

Basis	and	Dimension.
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Basis of a Vector Space

A set of vectors

$$\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$$

is a **basis** of a vector space if:

- 1 the vectors span the space, and
- 2 the vectors are linearly independent.

Therefore,

$$\text{Basis} = \text{Spanning set} + \text{Linear independence}$$

A basis provides a set of fundamental directions from which every vector in the space can be constructed.

Standard Basis

The standard basis of $\mathbb{R}^2$ is

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}.$$

Every vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}$$

can be written as

$$\mathbf{x} = x_1 \begin{bmatrix} 1 \\ 0 \end{bmatrix} + x_2 \begin{bmatrix} 0 \\ 1 \end{bmatrix}.$$

Thus, the basis vectors provide the coordinate directions of $\mathbb{R}^2$.

Standard Basis of $\mathbb{R}^3$

The standard basis of $\mathbb{R}^3$ is

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}.$$

Any vector

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}$$

can be expressed as

$$\mathbf{x} = x_1 \mathbf{e}_1 + x_2 \mathbf{e}_2 + x_3 \mathbf{e}_3.$$

The three basis vectors provide three independent directions.

Coordinates Relative to a Basis

Let

$$\mathcal{B} = \{\mathbf{v}_1, \mathbf{v}_2\}$$

be a basis of $\mathbb{R}^2$.

A vector $\mathbf{x}$ can be expressed as

$$\mathbf{x} = c_1 \mathbf{v}_1 + c_2 \mathbf{v}_2.$$

The coefficients form the **coordinate vector** of $\mathbf{x}$ relative to $\mathcal{B}$:

$$\boxed{[\mathbf{x}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} .}$$

The same geometric vector can therefore have different coordinates when different bases are used.

Dimension

The **dimension** of a vector space is the number of vectors in any basis of that space.
For example,

$$\dim(\mathbb{R}^2) = 2$$

and

$$\dim(\mathbb{R}^3) = 3.$$

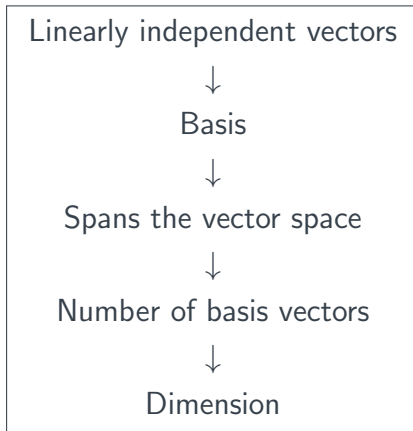
More generally,

$$\dim(\mathbb{R}^n) = n.$$

Dimension tells us how many independent directions are needed to describe the entire space.

Basis, Span and Dimension

These concepts are closely connected.



A basis is therefore a **minimal set of generators** for a vector space.

What Is a Vector Space?

A **vector space** is a set of objects that is closed under:

- ▶ vector addition
- ▶ scalar multiplication

and satisfies a collection of algebraic rules.

If $\mathbf{u}$ and $\mathbf{v}$ belong to a vector space V , and c is a scalar, then

$$\mathbf{u} + \mathbf{v} \in V$$

and

$$c\mathbf{u} \in V.$$

The scalars are usually elements of $\mathbb{R}$ or $\mathbb{C}$.

Vector Space Properties

For vectors $\mathbf{u}, \mathbf{v}, \mathbf{w} \in V$ and scalar c , the vector-space operations satisfy properties such as:

- ▶ $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- ▶ $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$
- ▶ There exists a zero vector $\mathbf{0}$
- ▶ Every vector has an additive inverse
- ▶ $c(\mathbf{u} + \mathbf{v}) = c\mathbf{u} + c\mathbf{v}$
- ▶ $(c + d)\mathbf{u} = c\mathbf{u} + d\mathbf{u}$

These properties provide the algebraic structure required for vector-space operations.

Examples of Vector Spaces

Vector spaces are not limited to arrows in $\mathbb{R}^n$.
Examples include:

$\mathbb{R}^n$

$$\begin{bmatrix} x_1 \\ \vdots \\ x_n \end{bmatrix}$$

Polynomials

$$p(x) = a_0 + a_1x + \cdots + a_nx^n$$

Matrices

Matrices of a fixed size form a vector space under matrix addition and scalar multiplication.

Functions

Suitable sets of functions can also form vector spaces.

Scalars and Fields

A **scalar** is a quantity used to multiply vectors.

Common scalar sets include

$$\mathbb{R} \quad \text{and} \quad \mathbb{C}.$$

A **field** provides the arithmetic structure required for scalar operations.

For example, the real numbers $\mathbb{R}$ form a field with

- ▶ addition
- ▶ subtraction
- ▶ multiplication
- ▶ division by non-zero numbers

Vector spaces are therefore defined relative to a chosen field of scalars.

Algebraic Structures

The lecture material introduces increasingly structured algebraic systems.



Each step adds additional algebraic properties.

Semigroup and Monoid

A **semigroup** is a set with an associative binary operation.

$$(a * b) * c = a * (b * c).$$

A **monoid** is a semigroup that additionally contains an identity element e :

$$e * a = a * e = a.$$

These structures provide a foundation for understanding more structured algebraic systems.

Group and Abelian Group

A **group** is a monoid in which every element has an inverse.
For every a , there exists a^{-1} such that

$$aa^{-1} = a^{-1}a = e.$$

An **Abelian group** additionally satisfies commutativity:

$$a * b = b * a.$$

Vector addition forms an Abelian group.

From Algebraic Structure to Vector Spaces

The vector-space structure combines two types of operations:

Vector Addition

$$u + v$$

Vectors form an Abelian group under addition.

Scalar Multiplication

$$c v$$

Scalars act on vectors while satisfying compatibility rules.

Together, these operations produce the structure of a **vector space**.

Vector Spaces in Computing

The idea of a vector space allows many different objects to be treated using the same mathematical framework.

- ▶ Feature vectors in machine learning
- ▶ Coordinate vectors in computer graphics
- ▶ Matrix spaces in numerical computing
- ▶ Polynomial representations
- ▶ Function spaces in mathematical modelling

Different objects $\longrightarrow$ Common linear structure

Matrices

A matrix is a rectangular array of numbers.

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}.$$

A matrix with m rows and n columns has dimensions

$$\boxed{m \times n}.$$

Matrices can represent:

- ▶ systems of linear equations
- ▶ transformations
- ▶ datasets
- ▶ relationships between variables

Matrix–Vector Multiplication

Let

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}.$$

Then

$$A\mathbf{x} = \begin{bmatrix} a_{11}x_1 + a_{12}x_2 \\ a_{21}x_1 + a_{22}x_2 \end{bmatrix}.$$

Equivalently,

$$A\mathbf{x} = x_1 \begin{bmatrix} a_{11} \\ a_{21} \end{bmatrix} + x_2 \begin{bmatrix} a_{12} \\ a_{22} \end{bmatrix}$$

So matrix–vector multiplication is a linear combination of the columns of A .

A Matrix as a Transformation

A matrix can define a function

$$T : \mathbb{R}^n \rightarrow \mathbb{R}^m$$

through

$$T(\mathbf{x}) = A\mathbf{x}.$$

The matrix A determines how the input vector is transformed.

$$\mathbf{x} \xrightarrow{A} A\mathbf{x}$$

If

$$A \in \mathbb{R}^{m \times n},$$

then

$$\mathbf{x} \in \mathbb{R}^n \implies A\mathbf{x} \in \mathbb{R}^m.$$

Linear Transformations

A transformation T is linear if, for all vectors $\mathbf{u}$, $\mathbf{v}$ and scalars c ,

$$T(\mathbf{u} + \mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$$

and

$$T(c\mathbf{u}) = cT(\mathbf{u}).$$

These two properties can be combined as

$$T(c_1\mathbf{u} + c_2\mathbf{v}) = c_1T(\mathbf{u}) + c_2T(\mathbf{v}).$$

Example: Scaling Transformation

Consider

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

For

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix},$$

we obtain

$$A\mathbf{x} = \begin{bmatrix} 2x \\ 3y \end{bmatrix}.$$

Therefore, the transformation

$$T(\mathbf{x}) = A\mathbf{x}$$

scales the x -coordinate by 2 and the y -coordinate by 3.

Identity Transformation

The identity matrix is

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}.$$

For any vector $\mathbf{x}$,

$$\boxed{I\mathbf{x} = \mathbf{x}.$$

The corresponding transformation

$$T(\mathbf{x}) = I\mathbf{x}$$

leaves every vector unchanged.

Thus, the identity transformation acts as the **neutral transformation** under composition.

Rotation as a Linear Transformation

A rotation through angle θ in $\mathbb{R}^2$ is represented by

$$R_\theta = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

Therefore,

$$T(\mathbf{x}) = R_\theta \mathbf{x}.$$

For $\theta = 90^\circ$,

$$R_{90^\circ} = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}.$$

The transformation rotates every vector by 90° .

Reflection as a Linear Transformation

Reflection about the x -axis can be represented by

$$A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

For

$$\mathbf{x} = \begin{bmatrix} x \\ y \end{bmatrix},$$

we obtain

$$A\mathbf{x} = \begin{bmatrix} x \\ -y \end{bmatrix}.$$

The x -coordinate remains unchanged while the y -coordinate changes sign.

Columns Determine the Transformation

Let

$$A = \begin{bmatrix} | & | \\ \mathbf{a}_1 & \mathbf{a}_2 \\ | & | \end{bmatrix}.$$

For

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix},$$

we have

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2.$$

In particular,

$$A \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \mathbf{a}_1, \quad A \begin{bmatrix} 0 \\ 1 \end{bmatrix} = \mathbf{a}_2.$$

Thus, the columns of a matrix are the images of the standard basis vectors.

Composition of Transformations

Suppose two linear transformations are represented by

$$T(\mathbf{x}) = A\mathbf{x}$$

and

$$S(\mathbf{x}) = B\mathbf{x}.$$

Applying T first and then S gives

$$S(T(\mathbf{x})) = B(A\mathbf{x}).$$

Therefore,

$$\boxed{S \circ T \longleftrightarrow BA.}$$

Matrix multiplication therefore provides a compact way to represent the **composition of linear transformations**.

Order of Composition Matters

In general,

$$AB \neq BA.$$

Therefore,

$$B(A\mathbf{x}) \neq A(B\mathbf{x})$$

in general.

This means that applying transformation A followed by B is generally different from applying B followed by A .

Key Idea

Matrix multiplication is generally **not commutative**.

Composition: Example

Consider

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad B = \begin{bmatrix} 1 & 0 \\ 0 & 3 \end{bmatrix}.$$

Then

$$BA = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}.$$

For this particular pair,

$$AB = BA.$$

But this is not generally true.

For transformations such as rotations, reflections, and shears, the order can produce different results.

Three-Dimensional Linear Transformations

A linear transformation in three dimensions has the form

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

with

$$T(\mathbf{x}) = A\mathbf{x},$$

where

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}.$$

The matrix can represent transformations such as:

- ▶ scaling
- ▶ rotation
- ▶ reflection
- ▶ shear

3D Scaling

A three-dimensional scaling transformation can be written as

$$A = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{bmatrix}.$$

Then

$$\begin{bmatrix} x' \\ y' \\ z' \end{bmatrix} = A \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

gives

$$\boxed{x' = s_x x, \quad y' = s_y y, \quad z' = s_z z.}$$

Different scaling factors can independently stretch or compress the three coordinate directions.

Rotation About the z-Axis

A rotation through θ about the z-axis is represented by

$$R_z(\theta) = \begin{bmatrix} \cos \theta & -\sin \theta & 0 \\ \sin \theta & \cos \theta & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

The x and y coordinates rotate in their plane while the z coordinate remains unchanged.

This type of transformation is fundamental in

3D graphics and computer vision.

Transformation Pipeline

Multiple transformations can be combined into a single matrix.

$$\mathbf{x} \xrightarrow{A} A\mathbf{x} \xrightarrow{B} BA\mathbf{x} \xrightarrow{C} CBA\mathbf{x}.$$

Therefore,

$$T(\mathbf{x}) = CBA\mathbf{x}.$$

This idea is central to transformation pipelines in:

- ▶ computer graphics
- ▶ robotics
- ▶ computer vision
- ▶ geometric modelling

Beyond Linear Transformations

Pure linear transformations preserve the origin:

$$\boxed{T(\mathbf{0}) = \mathbf{0}.$$

Translations do not satisfy this property.

In computer graphics, translations are commonly incorporated using **homogeneous coordinates**.

For example,

$$\begin{bmatrix} x' \\ y' \\ 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & t_x \\ 0 & 1 & t_y \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ 1 \end{bmatrix}.$$

This allows translation and linear transformations to be combined into matrix operations.

System of Linear Equations

A **linear equation** in variables $x_1, x_2, \dots, x_n$ has the form

$$a_1x_1 + a_2x_2 + \dots + a_nx_n = b.$$

A **system of linear equations** consists of multiple linear equations considered simultaneously.

For example,

$$\begin{cases} 2x + y = 5, \\ x - y = 1. \end{cases}$$

The objective is to find values of the variables that satisfy **all equations simultaneously**.

Solution of a System

For the system

$$\begin{cases} 2x + y = 5, \\ x - y = 1, \end{cases}$$

adding the two equations gives

$$3x = 6.$$

Therefore,

$$x = 2.$$

Substituting into

$$x - y = 1$$

gives

$$y = 1.$$

Hence,

$$(x, y) = (2, 1)$$

is the solution.

Geometric Interpretation

In two variables, each linear equation represents a line.

For example,

$$2x + y = 5$$

and

$$x - y = 1$$

represent two lines in the plane.

Their solution is the point where the two lines intersect.

Solution of system = Common intersection
--

This geometric interpretation becomes more general when we move to higher dimensions.

Consistency of a System

A system of linear equations is **consistent** if it has at least one solution.
It is **inconsistent** if it has no solution.

A consistent system may have:

Exactly one solution

or

Infinitely many solutions.

Therefore, the three possibilities are:

Unique solution

Infinitely many solutions

No solution

Consistency: Geometric View

For two equations in two variables:

One Solution

Two distinct lines intersect at one point.

No Solution

Parallel distinct lines never intersect.

Infinite Solutions

Coincident lines represent the same equation.

Thus, consistency is closely related to the geometry of the equations.

Matrix Form of a Linear System

Consider

$$\begin{cases} a_{11}x_1 + a_{12}x_2 = b_1, \\ a_{21}x_1 + a_{22}x_2 = b_2. \end{cases}$$

Define

$$A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}, \quad \mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \end{bmatrix}, \quad \mathbf{b} = \begin{bmatrix} b_1 \\ b_2 \end{bmatrix}.$$

Then the system becomes

$$\boxed{A\mathbf{x} = \mathbf{b}.}$$

The Equation $Ax = b$

The matrix equation

$$Ax = b$$

can be interpreted as a transformation.

$$x \xrightarrow{A} Ax = b.$$

The problem is therefore:

Find the input x that produces the output b .

This connects systems of equations directly with **linear transformations**.

$Ax = b$ as a Linear Combination

Let

$$A = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & \cdots & | \end{bmatrix},$$

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

Then

$$A\mathbf{x} = x_1\mathbf{a}_1 + x_2\mathbf{a}_2 + \cdots + x_n\mathbf{a}_n.$$

Therefore,

$$A\mathbf{x} = \mathbf{b}$$

asks whether $\mathbf{b}$ can be expressed as a linear combination of the columns of A .

$$\mathbf{b} \in \text{Col}(A)$$

This is precisely the condition for consistency.

Augmented Matrix

A system can be represented using an **augmented matrix**.

For

$$\begin{cases} 2x + y = 5, \\ x - y = 1, \end{cases}$$

the augmented matrix is

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & -1 & 1 \end{array} \right].$$

The vertical line separates:

coefficient matrix from right-hand side.

Elementary Row Operations

Gaussian elimination uses three elementary row operations.

- 1 Interchange two rows.
- 2 Multiply a row by a non-zero scalar.
- 3 Add a multiple of one row to another row.

These operations transform the system into an equivalent system with the same solution set.

Equivalent system $\implies$ Same solutions

Gaussian Elimination

Gaussian elimination transforms an augmented matrix into **row echelon form**.

Consider

$$\left[\begin{array}{cc|c} 2 & 1 & 5 \\ 1 & -1 & 1 \end{array} \right].$$

Swap the rows:

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 2 & 1 & 5 \end{array} \right].$$

Then perform

$$R_2 \leftarrow R_2 - 2R_1.$$

This gives

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 3 & 3 \end{array} \right].$$

Gaussian Elimination: Back Substitution

From

$$\left[\begin{array}{cc|c} 1 & -1 & 1 \\ 0 & 3 & 3 \end{array} \right]$$

we obtain

$$3y = 3$$

and therefore

$$y = 1.$$

Substituting into the first equation,

$$x - y = 1$$

gives

$$x = 2.$$

Hence,

$$\mathbf{x} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Row Echelon Form

A matrix is in **row echelon form** when:

- ▶ all non-zero rows occur above zero rows,
- ▶ each leading entry is to the right of the leading entry in the row above it,
- ▶ entries below each leading entry are zero.

For example,

$$\begin{bmatrix} 1 & 2 & -1 \\ 0 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}$$

is in row echelon form.

Gaussian elimination systematically produces this structure.

Reduced Row Echelon Form

A matrix is in **reduced row echelon form** when, in addition to row echelon conditions:

- ▶ every leading entry is 1,
- ▶ each leading 1 is the only non-zero entry in its column.

For example,

$$\begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & -3 \\ 0 & 0 & 0 \end{bmatrix}$$

is in reduced row echelon form.

RREF makes the solution structure of a system easier to identify.

Identifying a Unique Solution

Consider

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 1 & 2 \end{array} \right].$$

Every variable has a pivot.

Therefore, the system has

exactly one solution.

In general, a unique solution occurs when every variable is determined by the equations.

Detecting No Solution

Suppose elimination produces

$$\left[\begin{array}{ccc|c} 1 & 2 & 0 & 3 \\ 0 & 0 & 0 & 5 \end{array} \right].$$

The second row represents

$$0 = 5.$$

This is impossible.

Therefore,

The system is inconsistent.

A row of the form

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & c \end{array} \right], \quad c \neq 0,$$

signals that the system has **no solution**.

Infinitely Many Solutions

Consider a system whose reduced form contains

$$\left[\begin{array}{ccc|c} 1 & 0 & 2 & 4 \\ 0 & 1 & -1 & 3 \\ 0 & 0 & 0 & 0 \end{array} \right].$$

The third row gives no new constraint. One variable can therefore be chosen freely.
Let

$$z = t.$$

Then

$$x = 4 - 2t, \quad y = 3 + t.$$

Hence,

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 3 \\ 0 \end{bmatrix} + t \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}.$$

There are infinitely many solutions.

Solving $Ax = b$ Using an Inverse

Suppose A is a square matrix with an inverse.
Starting with

$$Ax = b,$$

multiply both sides by A^{-1} :

$$A^{-1}Ax = A^{-1}b.$$

Since

$$A^{-1}A = I,$$

we obtain

$$\boxed{x = A^{-1}b.}$$

This gives a direct formula for the solution.

When Does $A^{-1}b$ Exist?

The expression

$$x = A^{-1}b$$

requires A to be:

- ▶ square,
- ▶ invertible.

For a square matrix,

$$A^{-1} \text{ exists} \iff \det(A) \neq 0.$$

When these conditions hold, the system

$$Ax = b$$

has a unique solution for every b .

Two Ways to Solve a Linear System

For a system

$$Ax = b,$$

two approaches are:

Gaussian Elimination

Transform the augmented matrix and solve the resulting system.

$$[A \mid b]$$

Inverse Method

If A is invertible,

$$x = A^{-1}b.$$

For computational problems involving many right-hand sides, factorization-based methods are generally more useful than explicitly computing A^{-1} .

Homogeneous Linear Systems

A homogeneous system has the form

$$Ax = \mathbf{0}.$$

It always has at least the trivial solution

$$\mathbf{x} = \mathbf{0}.$$

Depending on A , there may also be non-zero solutions.

Trivial solution	vs.	Non-trivial solutions
------------------	-----	-----------------------

This will become important when studying the **null space** and linear dependence.

The Determinant

The **determinant** is a scalar associated with a square matrix.

For a 2×2 matrix

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is

$$\det(A) = ad - bc.$$

The determinant provides information about the transformation represented by the matrix.

It also plays a central role in determining whether a matrix is invertible.

Example: Determinant of a 2×2 Matrix

Consider

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}.$$

Then

$$\det(A) = (3)(4) - (2)(1).$$

Therefore,

$$\det(A) = 10.$$

Since

$$\det(A) \neq 0,$$

the matrix is invertible.

Geometric Meaning of the Determinant

For a 2×2 matrix, the absolute value

$$|\det(A)|$$

represents the factor by which the transformation changes **area**.

If

$$|\det(A)| > 1,$$

areas are enlarged.

If

$$0 < |\det(A)| < 1,$$

areas are reduced.

If

$$\boxed{\det(A) = 0},$$

the transformation collapses the plane into a lower-dimensional space.

Determinant in Three Dimensions

For a 3×3 matrix, the absolute value of the determinant represents the volume-scaling factor.

$$|\det(A)| = \text{volume scaling factor}$$

For example,

$$\det(A) = 2$$

means that the transformation doubles volumes.

A negative determinant indicates that the transformation also reverses orientation.

Properties of the Determinant

For square matrices A and B :

$$\det(AB) = \det(A) \det(B)$$

and

$$\det(A^T) = \det(A).$$

For the identity matrix,

$$\det(I) = 1.$$

For a scalar c and an $n \times n$ matrix,

$$\det(cA) = c^n \det(A).$$

Singular and Non-Singular Matrices

A square matrix A is called **singular** if

$$\det(A) = 0.$$

A square matrix is **non-singular** if

$$\det(A) \neq 0.$$

Geometrically, a singular transformation collapses space into a lower-dimensional set.

For example, a plane may be collapsed into a line.

Determinant and Invertibility

For a square matrix A ,

$$A^{-1} \text{ exists} \iff \det(A) \neq 0.$$

Equivalently,

$$\det(A) = 0 \iff A \text{ is singular.}$$

Therefore, the determinant provides a quick test for whether a square matrix has an inverse.

This connects three ideas:

$$\text{Determinant} \iff \text{Invertibility} \iff \text{Unique solution}$$

Inverse Matrix

For a square matrix A , its inverse A^{-1} satisfies

$$AA^{-1} = A^{-1}A = I.$$

Therefore,

$$A^{-1}$$

undoes the transformation performed by A .

If

$$y = Ax,$$

then

$$x = A^{-1}y.$$

Thus, the inverse transformation recovers the original vector.

Inverse of a 2×2 Matrix

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

if

$$ad - bc \neq 0,$$

then

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

The denominator is precisely

$$\det(A).$$

Therefore,

$$\det(A) \neq 0$$

is required for the inverse to exist.

Example: Computing an Inverse

Consider

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

First,

$$\det(A) = (2)(1) - (1)(1) = 1.$$

Therefore,

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

Indeed,

$$AA^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

Inverse Matrix and Linear Systems

Consider

$$Ax = b.$$

If A is invertible,

$$A^{-1}Ax = A^{-1}b.$$

Therefore,

$$\boxed{x = A^{-1}b.}$$

Hence, an invertible coefficient matrix guarantees a unique solution for every right-hand side b .

Cofactors

For larger matrices, determinants can be computed using **cofactors**.

For an element a_{ij} ,

$$C_{ij} = (-1)^{i+j} M_{ij},$$

where M_{ij} is the determinant obtained after deleting row i and column j .

For a 3×3 matrix,

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}.$$

This is called **cofactor expansion**.

Adjugate and the Inverse

The matrix of cofactors is called the **cofactor matrix**.
Its transpose is the **adjugate**:

$$\text{adj}(A) = C^T.$$

For an invertible matrix,

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

This provides a general formula for the inverse of a square matrix.

For large matrices, however, direct inverse computation is usually not the preferred computational approach.

Cramer's Rule

Consider the system

$$Ax = b,$$

where A is an invertible $n \times n$ matrix.

Cramer's rule gives each component of the solution as

$$x_i = \frac{\det(A_i)}{\det(A)},$$

where A_i is obtained by replacing the i -th column of A with b .

The rule requires

$$\det(A) \neq 0.$$

Cramer's Rule for a 2×2 System

Consider

$$\begin{cases} a_1x + b_1y = c_1, \\ a_2x + b_2y = c_2. \end{cases}$$

The coefficient determinant is

$$D = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix}.$$

For x ,

$$D_x = \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix}.$$

For y ,

$$D_y = \begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix}.$$

Therefore,

$$\boxed{x = \frac{D_x}{D}, \quad y = \frac{D_y}{D}.$$

Example: Cramer's Rule

Consider

$$\begin{cases} 2x + y = 5, \\ x - y = 1. \end{cases}$$

The coefficient determinant is

$$D = \begin{vmatrix} 2 & 1 \\ 1 & -1 \end{vmatrix} = -3.$$

For x ,

$$D_x = \begin{vmatrix} 5 & 1 \\ 1 & -1 \end{vmatrix} = -6.$$

For y ,

$$D_y = \begin{vmatrix} 2 & 5 \\ 1 & 1 \end{vmatrix} = -3.$$

Hence,

$$\boxed{x = 2, \quad y = 1.}$$

Cramer's Rule and Gaussian Elimination

Both methods can solve

$$Ax = b.$$

Cramer's Rule

Uses determinants:

$$x_i = \frac{\det(A_i)}{\det(A)}.$$

Useful for theoretical derivations and small systems.

Gaussian Elimination

Uses elementary row operations.
Efficient and widely used for numerical computation.

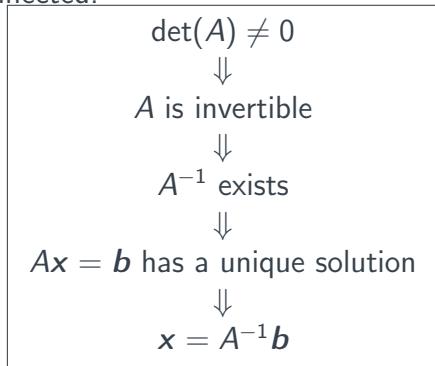
For large systems, repeatedly computing determinants is computationally expensive.

The Linear Algebra Connection

For a square system

$$Ax = b,$$

the following ideas are connected:



This connection will be extended further through **column space, null space, rank, and linear independence**.

The Determinant

The **determinant** is a scalar associated with a square matrix.

For a 2×2 matrix,

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

the determinant is

$$\det(A) = ad - bc$$

The determinant provides information about:

- ▶ whether a matrix is invertible,
- ▶ how a transformation changes area or volume,
- ▶ whether dimensions are collapsed by a transformation.

Example: Determinant of a 2×2 Matrix

Consider

$$A = \begin{bmatrix} 3 & 2 \\ 1 & 4 \end{bmatrix}.$$

Using

$$\det(A) = ad - bc,$$

we obtain

$$\det(A) = (3)(4) - (2)(1) = 12 - 2.$$

Therefore,

$$\det(A) = 10$$

Since the determinant is non-zero, A is invertible.

Geometric Meaning of the Determinant

For a 2×2 transformation,

$$|\det(A)|$$

represents the factor by which areas are scaled.

Determinant	Geometric effect
$ \det(A) > 1$	Area increases
$0 < \det(A) < 1$	Area decreases
$ \det(A) = 1$	Area is preserved
$\det(A) = 0$	Area collapses to zero

The sign of the determinant also indicates whether orientation is preserved or reversed.

Determinant in Three Dimensions

For a 3×3 matrix, the absolute value of the determinant represents the factor by which volumes are scaled.

$$|\det(A)| = \text{volume scaling factor}$$

For example,

$$\det(A) = 3$$

means that the transformation scales volume by a factor of 3.

A negative determinant indicates a reversal of orientation.

Properties of the Determinant

For square matrices A and B ,

$$\det(AB) = \det(A) \det(B)$$

and

$$\det(A^T) = \det(A).$$

For the identity matrix,

$$\det(I) = 1.$$

If A is invertible,

$$\det(A^{-1}) = \frac{1}{\det(A)}.$$

Singular and Non-Singular Matrices

A square matrix A is **singular** when

$$\det(A) = 0.$$

A square matrix is **non-singular** when

$$\det(A) \neq 0.$$

Geometrically, a singular transformation collapses the space into a lower-dimensional space.

For example,

$$\mathbb{R}^2 \longrightarrow \text{a line.}$$

Determinant and Invertibility

For a square matrix A ,

$$A^{-1} \text{ exists} \iff \det(A) \neq 0.$$

Therefore,

$$\det(A) = 0 \iff A \text{ is singular.}$$

This gives an important connection:

$$\text{Non-zero determinant} \rightarrow \text{Invertible matrix} \rightarrow \text{Inverse transformation}$$

Inverse Matrix

The inverse of a square matrix A is a matrix A^{-1} such that

$$AA^{-1} = A^{-1}A = I.$$

The inverse matrix represents the transformation that **undoes** the original transformation.

If

$$y = Ax,$$

then

$$x = A^{-1}y.$$

Thus, A^{-1} maps the output back to the original input.

Inverse of a 2×2 Matrix

For

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix},$$

if

$$ad - bc \neq 0,$$

then

$$A^{-1} = \frac{1}{ad - bc} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}.$$

Notice that

$$ad - bc = \det(A).$$

Therefore, the inverse exists precisely when

$$\det(A) \neq 0.$$

Example: Finding an Inverse

Let

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 1 \end{bmatrix}.$$

First,

$$\det(A) = (2)(1) - (1)(1) = 1.$$

Therefore,

$$A^{-1} = \begin{bmatrix} 1 & -1 \\ -1 & 2 \end{bmatrix}.$$

Verification:

$$AA^{-1} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

Inverse Matrix and Linear Systems

Consider

$$Ax = b.$$

If A is invertible, multiply both sides by A^{-1} :

$$A^{-1}Ax = A^{-1}b.$$

Since

$$A^{-1}A = I,$$

we obtain

$$\boxed{x = A^{-1}b.}$$

Thus, an invertible coefficient matrix gives a unique solution for every b .

Minors and Cofactors

For an element a_{ij} of a matrix, its **minor** M_{ij} is the determinant obtained by deleting row i and column j .

The corresponding cofactor is

$$C_{ij} = (-1)^{i+j} M_{ij}.$$

The signs follow the pattern

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}.$$

Cofactors can be used to calculate determinants of larger matrices.

Cofactor Expansion

For

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix},$$

expanding along the first row gives

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}.$$

Equivalently,

$$\det(A) = a_{11}M_{11} - a_{12}M_{12} + a_{13}M_{13}.$$

The same determinant can be expanded along any row or column.

Adjugate Matrix

The matrix of cofactors is

$$C = \begin{bmatrix} C_{11} & C_{12} & \cdots \\ C_{21} & C_{22} & \cdots \\ \vdots & \vdots & \ddots \end{bmatrix}.$$

The **adjugate** is the transpose of the cofactor matrix:

$$\text{adj}(A) = C^T.$$

For an invertible matrix,

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A).$$

This provides a general formula for computing the inverse.

Computing Inverses in Practice

Although

$$A^{-1} = \frac{1}{\det(A)} \operatorname{adj}(A)$$

is mathematically useful, it is not usually the preferred method for large numerical problems.

For computational applications, methods such as

- ▶ Gaussian elimination,
- ▶ LU factorization,
- ▶ QR factorization

are generally more practical.

The inverse formula is therefore particularly useful for understanding the mathematics and for small matrices.

Column Space

Let

$$A = \left[\begin{array}{c|c|c|c} | & | & & | \\ \mathbf{a}_1 & \mathbf{a}_2 & \cdots & \mathbf{a}_n \\ | & | & & | \end{array} \right] .$$

The **column space** of A is the set of all linear combinations of its columns.

$$\text{Col}(A) = \text{span}\{\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n\}$$

Therefore, the column space contains every vector that can be obtained in the form

$$A\mathbf{x}.$$

Column Space and $Ax = b$

Consider

$$Ax = b.$$

Since

$$Ax = x_1 \mathbf{a}_1 + x_2 \mathbf{a}_2 + \cdots + x_n \mathbf{a}_n,$$

a solution exists precisely when $\mathbf{b}$ can be expressed as a linear combination of the columns of A .

Therefore,

$$Ax = \mathbf{b} \text{ is consistent} \iff \mathbf{b} \in \text{Col}(A).$$

The column space therefore describes all possible outputs of the transformation represented by A .

Example: Column Space

Consider

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}.$$

Its columns are

$$\mathbf{a}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{a}_2 = \begin{bmatrix} 2 \\ 4 \end{bmatrix}.$$

Since

$$\mathbf{a}_2 = 2\mathbf{a}_1,$$

the two columns do not generate the entire plane.

Thus,

$$\text{Col}(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}.$$

The column space is a line through the origin.

Null Space

The **null space** of a matrix A is the set of all vectors $\mathbf{x}$ satisfying

$$A\mathbf{x} = \mathbf{0}.$$

It is denoted by

$$\text{Null}(A) = \{\mathbf{x} : A\mathbf{x} = \mathbf{0}\}.$$

The null space contains all inputs that are mapped to the zero vector.

$$\mathbf{x} \xrightarrow{A} \mathbf{0}.$$

Example: Null Space

Consider

$$A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$$

$$A \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$$

Solving the system gives

$$x + 2y = 0 \quad \implies \quad x = -2y.$$

Let $y = t$. Then

$$\begin{bmatrix} x \\ y \end{bmatrix} = t \begin{bmatrix} -2 \\ 1 \end{bmatrix}.$$

Therefore,

$$\text{Null}(A) = \text{span} \left\{ \begin{bmatrix} -2 \\ 1 \end{bmatrix} \right\}$$

Null Space as the Kernel

For a linear transformation

$$T(\mathbf{x}) = A\mathbf{x},$$

the null space is also called the **kernel**.

$$\text{Null}(A) = \ker(T)$$

It contains all vectors that are mapped to zero:

$$\ker(T) = \{\mathbf{x} : T(\mathbf{x}) = \mathbf{0}\}.$$

The kernel measures which directions are lost or collapsed by the transformation.

Basis for the Column Space

A basis for the column space consists of linearly independent columns that span the column space.

Consider

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}.$$

Since

$$\mathbf{a}_2 = 2\mathbf{a}_1, \quad \mathbf{a}_3 = 3\mathbf{a}_1,$$

only the first column is needed.

Therefore,

$$\left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix} \right\}$$

is a basis for $\text{Col}(A)$.

Pivot Columns and Column Space

Gaussian elimination can be used to identify independent columns.
Consider

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}.$$

Row reduction gives

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 0 & 0 \end{bmatrix}.$$

There is one pivot.

The corresponding column of the **original matrix** forms a basis for the column space.

$\text{rank}(A) = \text{number of pivot columns.}$

Rank of a Matrix

The **rank** of a matrix is the dimension of its column space.

$$\text{rank}(A) = \dim(\text{Col}(A)).$$

Equivalently, the rank is the number of pivot columns obtained during row reduction.

For an $m \times n$ matrix,

$$\text{rank}(A) \leq \min(m, n).$$

Rank measures the number of independent directions produced by the transformation.

Row Space

The **row space** of a matrix is the span of its rows.

For

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix},$$

the rows are

$$\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}, \quad \begin{bmatrix} 2 & 4 & 6 \end{bmatrix}.$$

Since the second row is twice the first,

$$\text{Row}(A) = \text{span} \left\{ \begin{bmatrix} 1 & 2 & 3 \end{bmatrix} \right\}.$$

The dimension of the row space is also the rank of A .

Nullity

The **nullity** of a matrix is the dimension of its null space.

$$\text{nullity}(A) = \dim(\text{Null}(A)).$$

For an $m \times n$ matrix,

$$\text{rank}(A) + \text{nullity}(A) = n.$$

This is the **Rank–Nullity Theorem**.

The value n is the number of columns of A .

Example: Rank–Nullity

Consider

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 6 \end{bmatrix}.$$

There are

$$n = 3$$

columns.

From row reduction,

$$\text{rank}(A) = 1.$$

Therefore,

$$1 + \text{nullity}(A) = 3.$$

Hence,

$$\boxed{\text{nullity}(A) = 2.}$$

So the matrix has one independent output direction and two independent null directions.

Linear Dependence and the Null Space

Suppose the columns of A are

$$\mathbf{a}_1, \mathbf{a}_2, \dots, \mathbf{a}_n.$$

A linear dependence relation has the form

$$c_1 \mathbf{a}_1 + c_2 \mathbf{a}_2 + \dots + c_n \mathbf{a}_n = \mathbf{0},$$

where at least one coefficient is non-zero.

But this can be written as

$$A \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix} = \mathbf{0}.$$

Therefore,

Non-trivial null vectors $\iff$ Linear dependence among columns.

Linear Independence and the Null Space

The columns of A are linearly independent if

$$A\mathbf{x} = \mathbf{0}$$

has only the trivial solution

$$\mathbf{x} = \mathbf{0}.$$

Thus,

$$\text{Null}(A) = \{\mathbf{0}\} \iff \text{columns of } A \text{ are linearly independent.}$$

A non-zero vector in the null space therefore provides a direct certificate of linear dependence.

Column Space vs. Null Space

For a matrix A :

Column Space

$$\text{Col}(A)$$

Contains all possible outputs

$$Ax.$$

Its dimension is the rank.

Null Space

$$\text{Null}(A)$$

Contains all inputs satisfying

$$Ax = \mathbf{0}.$$

Its dimension is the nullity.

One describes possible **outputs**; the other describes inputs that are **collapsed to zero**.

The Fundamental Connection

For

$$Ax = b,$$

two fundamental questions are:

$$b \in \text{Col}(A) ?$$

determines whether a solution exists.

If a solution x_p exists, then every solution has the form

$$x = x_p + x_n, \quad x_n \in \text{Null}(A).$$

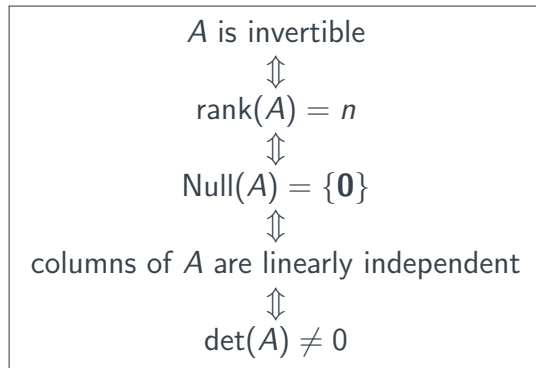
Therefore:

$$\text{Particular solution} + \text{Null-space solution}$$

describes the complete solution set.

Column Space, Null Space and Invertibility

For an $n \times n$ matrix A , the following are equivalent:



These equivalences connect the ideas developed throughout linear algebra.

Dot Product

For two vectors

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix},$$

their **dot product** is

$$\mathbf{u} \cdot \mathbf{v} = \sum_{i=1}^n u_i v_i$$

or equivalently,

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}.$$

The result is a **scalar**, not a vector.

Example: Dot Product

Let

$$\mathbf{u} = \begin{bmatrix} 2 \\ -1 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix}.$$

Then

$$\mathbf{u} \cdot \mathbf{v} = (2)(1) + (-1)(4) + (3)(2).$$

Therefore,

$$\boxed{\mathbf{u} \cdot \mathbf{v} = 4.}$$

Using matrix notation,

$$\mathbf{u}^T \mathbf{v} = \begin{bmatrix} 2 & -1 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 4 \\ 2 \end{bmatrix} = 4.$$

Geometric Meaning of the Dot Product

For non-zero vectors,

$$\mathbf{u} \cdot \mathbf{v} = \|\mathbf{u}\| \|\mathbf{v}\| \cos \theta$$

where θ is the angle between the vectors.

Therefore,

$$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{u}\| \|\mathbf{v}\|}$$

The dot product therefore measures the directional relationship between two vectors.

Orthogonality

Two vectors are **orthogonal** if they are perpendicular.

For non-zero vectors,

$$\mathbf{u} \perp \mathbf{v}$$

precisely when

$$\boxed{\mathbf{u} \cdot \mathbf{v} = 0.}$$

For example,

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 2 \\ -1 \end{bmatrix}.$$

Then

$$\mathbf{u} \cdot \mathbf{v} = (1)(2) + (2)(-1) = 0.$$

Hence,

$$\boxed{\mathbf{u} \perp \mathbf{v}.$$

Projection of One Vector onto Another

The projection of $\mathbf{u}$ onto a non-zero vector $\mathbf{v}$ is

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\mathbf{v} \cdot \mathbf{v}} \mathbf{v}.$$

Since

$$\mathbf{v} \cdot \mathbf{v} = \|\mathbf{v}\|^2,$$

we may also write

$$\text{proj}_{\mathbf{v}} \mathbf{u} = \frac{\mathbf{u} \cdot \mathbf{v}}{\|\mathbf{v}\|^2} \mathbf{v}.$$

The result is the component of $\mathbf{u}$ in the direction of $\mathbf{v}$.

Orthogonal Decomposition

A vector $\mathbf{u}$ can be decomposed into two components:

$$\mathbf{u} = \mathbf{u}_{\parallel} + \mathbf{u}_{\perp}.$$

Here,

$$\mathbf{u}_{\parallel} = \text{proj}_{\mathbf{v}} \mathbf{u}$$

is parallel to $\mathbf{v}$, while

$$\mathbf{u}_{\perp} = \mathbf{u} - \mathbf{u}_{\parallel}$$

is perpendicular to $\mathbf{v}$.

Therefore,

$$\mathbf{u}_{\perp} \cdot \mathbf{v} = 0.$$

This decomposition is fundamental in geometry, optimization, and machine learning.

Dot Product and Duality

Fix a vector $\mathbf{u}$.

The mapping

$$f_{\mathbf{u}}(\mathbf{x}) = \mathbf{u} \cdot \mathbf{x}$$

takes a vector as input and produces a scalar.

It is linear because

$$f_{\mathbf{u}}(\alpha \mathbf{x} + \beta \mathbf{y}) = \alpha f_{\mathbf{u}}(\mathbf{x}) + \beta f_{\mathbf{u}}(\mathbf{y}).$$

Therefore,

$$\boxed{\mathbf{x} \mapsto \mathbf{u} \cdot \mathbf{x}}$$

is a **linear functional**.

Dual Space

The set of all linear functionals on a vector space V forms its **dual space**, denoted by

$$V^*.$$

A vector u can define a linear functional through

$$x \mapsto u^T x.$$

Thus, the dot product provides a natural connection between vectors and linear functionals.

$$\text{Vector} \longrightarrow \text{Linear functional}$$

This connection is known as **duality**.

Cross Product

For vectors

$$\mathbf{u}, \mathbf{v} \in \mathbb{R}^3,$$

the **cross product**

$$\mathbf{u} \times \mathbf{v}$$

produces another vector in $\mathbb{R}^3$.

If

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix},$$

then

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} u_2 v_3 - u_3 v_2 \\ u_3 v_1 - u_1 v_3 \\ u_1 v_2 - u_2 v_1 \end{bmatrix}.$$

Example: Cross Product

Let

$$\mathbf{u} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}.$$

Then

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} 2(6) - 3(5) \\ 3(4) - 1(6) \\ 1(5) - 2(4) \end{bmatrix}.$$

Therefore,

$$\mathbf{u} \times \mathbf{v} = \begin{bmatrix} -3 \\ 6 \\ -3 \end{bmatrix}.$$

Cross Product and Orthogonality

The cross product is perpendicular to both input vectors:

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{u} = 0$$

and

$$(\mathbf{u} \times \mathbf{v}) \cdot \mathbf{v} = 0.$$

Therefore,

$$\mathbf{u} \times \mathbf{v} \perp \mathbf{u}$$

and

$$\mathbf{u} \times \mathbf{v} \perp \mathbf{v}.$$

The three vectors therefore define a mutually related three-dimensional geometry.

Magnitude of the Cross Product

The magnitude of the cross product is

$$\|\mathbf{u} \times \mathbf{v}\| = \|\mathbf{u}\| \|\mathbf{v}\| \sin \theta.$$

This is equal to the area of the parallelogram formed by $\mathbf{u}$ and $\mathbf{v}$.

Therefore,

$$\text{Area} = \|\mathbf{u} \times \mathbf{v}\|.$$

If the vectors are parallel,

$$\theta = 0 \implies \mathbf{u} \times \mathbf{v} = \mathbf{0}.$$

Direction of the Cross Product

The direction of

$$\mathbf{u} \times \mathbf{v}$$

is determined by the **right-hand rule**.

The cross product is anti-commutative:

$$\mathbf{u} \times \mathbf{v} = -(\mathbf{v} \times \mathbf{u}).$$

Also,

$$\mathbf{u} \times \mathbf{u} = \mathbf{0}.$$

Thus, reversing the order reverses the direction of the resulting vector.

Cross Products of Standard Basis Vectors

For the standard basis vectors

$$\hat{i}, \quad \hat{j}, \quad \hat{k},$$

the fundamental relationships are

$$\hat{i} \times \hat{j} = \hat{k}$$

$$\hat{j} \times \hat{k} = \hat{i}$$

$$\hat{k} \times \hat{i} = \hat{j}.$$

Reversing the order changes the sign.

Cross Product as Matrix Multiplication

Fix

$$\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}.$$

Define the skew-symmetric matrix

$$[\mathbf{u}]_{\times} = \begin{bmatrix} 0 & -u_3 & u_2 \\ u_3 & 0 & -u_1 \\ -u_2 & u_1 & 0 \end{bmatrix}.$$

Then

$$\boxed{\mathbf{u} \times \mathbf{v} = [\mathbf{u}]_{\times} \mathbf{v} .}$$

Therefore, for fixed $\mathbf{u}$, the cross product is a **linear transformation** of $\mathbf{v}$.

Cross Product as a Linear Transformation

Fix $\mathbf{u} \in \mathbb{R}^3$ and define

$$T(\mathbf{v}) = \mathbf{u} \times \mathbf{v}.$$

Then

$$T(\alpha\mathbf{v} + \beta\mathbf{w}) = \alpha T(\mathbf{v}) + \beta T(\mathbf{w}).$$

Therefore,

$$T : \mathbb{R}^3 \rightarrow \mathbb{R}^3$$

is a linear transformation.

Its matrix representation is

$$T(\mathbf{v}) = [\mathbf{u}]_{\times} \mathbf{v}.$$

Kernel of the Cross Product Transformation

Consider

$$T(\mathbf{v}) = \mathbf{u} \times \mathbf{v}.$$

For $\mathbf{u} \neq \mathbf{0}$,

$$T(\mathbf{v}) = \mathbf{0}$$

when $\mathbf{v}$ is parallel to $\mathbf{u}$.

Therefore,

$$\text{Null}(T) = \text{span}\{\mathbf{u}\}.$$

The cross-product transformation therefore collapses one direction to zero.
This connects directly with the concept of **null space**.

Range of the Cross Product Transformation

For

$$T(\mathbf{v}) = \mathbf{u} \times \mathbf{v},$$

every output is perpendicular to $\mathbf{u}$.

Therefore,

$$\text{Range}(T) = \mathbf{u}^\perp.$$

The range is the plane through the origin perpendicular to $\mathbf{u}$.

Thus,

$$\dim(\text{Null}(T)) = 1$$

and

$$\dim(\text{Range}(T)) = 2.$$

This satisfies

$$1 + 2 = 3,$$

as predicted by the Rank–Nullity Theorem.

Dot Product vs. Cross Product

	Dot Product	Cross Product
Input	$\mathbb{R}^n$	$\mathbb{R}^3$
Output	Scalar	Vector
Notation	$\boldsymbol{u} \cdot \boldsymbol{v}$	$\boldsymbol{u} \times \boldsymbol{v}$
Geometric role	Angle / projection	Area / perpendicular direction
Zero condition	Orthogonal vectors	Parallel vectors

Both operations provide important geometric information about vectors.

Basis and Coordinates

A vector can be represented using different bases.

Consider the standard basis of $\mathbb{R}^2$:

$$\mathcal{E} = \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \end{bmatrix} \right\}.$$

For

$$\mathbf{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix},$$

its coordinates in the standard basis are

$$[\mathbf{v}]_{\mathcal{E}} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

The vector itself and its coordinate representation are related, but are not the same concept.

A Different Basis

Consider the basis

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

We want to express

$$\mathbf{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

using this basis.

Write

$$\mathbf{v} = c_1 \begin{bmatrix} 1 \\ 1 \end{bmatrix} + c_2 \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

This gives

$$c_1 + c_2 = 3, \quad c_1 - c_2 = 1.$$

Hence,

$$c_1 = 2, \quad c_2 = 1.$$

Therefore,

$$\boxed{[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix} .}$$

Vector vs. Coordinate Representation

For the same vector $\mathbf{v}$,

$$[\mathbf{v}]_{\mathcal{E}} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

while

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

The numerical coordinates are different because the bases are different.

However,

The vector itself has not changed.

Only its representation has changed.

Basis and Coordinates

Let

$$\mathcal{B} = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_n\}$$

be a basis for $\mathbb{R}^n$.

Every vector $\mathbf{v} \in \mathbb{R}^n$ can be written uniquely as

$$\mathbf{v} = c_1 \mathbf{b}_1 + c_2 \mathbf{b}_2 + \dots + c_n \mathbf{b}_n$$

The coefficients form the coordinate vector of $\mathbf{v}$ with respect to $\mathcal{B}$:

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} c_1 \\ c_2 \\ \vdots \\ c_n \end{bmatrix}$$

Basis Matrix

The basis vectors can be arranged as the columns of a matrix:

$$P_{\mathcal{B}} = \begin{bmatrix} | & | & \cdots & | \\ b_1 & b_2 & \cdots & b_n \\ | & | & \cdots & | \end{bmatrix}$$

Then the vector can be reconstructed from its coordinates:

$$\mathbf{v} = P_{\mathcal{B}}[\mathbf{v}]_{\mathcal{B}}$$

Interpretation

$P_{\mathcal{B}}$ converts coordinates expressed in basis $\mathcal{B}$ into the standard-coordinate representation.

From Basis Coordinates to Standard Coordinates

Consider the basis

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}.$$

Its basis matrix is

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

Suppose the coordinates of $\mathbf{v}$ in $\mathcal{B}$ are

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Then

$$\mathbf{v} = P_{\mathcal{B}}[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \end{bmatrix} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$$

From Standard Coordinates to Basis Coordinates

Starting from

$$\mathbf{v} = P_{\mathcal{B}}[\mathbf{v}]_{\mathcal{B}},$$

multiply by $P_{\mathcal{B}}^{-1}$:

$$P_{\mathcal{B}}^{-1}\mathbf{v} = [\mathbf{v}]_{\mathcal{B}}.$$

Therefore,

$$[\mathbf{v}]_{\mathcal{B}} = P_{\mathcal{B}}^{-1}\mathbf{v}.$$

Thus,

$$\begin{array}{ccc} [\mathbf{v}]_{\mathcal{B}} & \xrightarrow{P_{\mathcal{B}}} & \mathbf{v} \\ \mathbf{v} & \xrightarrow{P_{\mathcal{B}}^{-1}} & [\mathbf{v}]_{\mathcal{B}} \end{array}$$

Change of Basis

Suppose $\mathcal{B}$ and $\mathcal{C}$ are two bases.

We can convert coordinates from $\mathcal{B}$ to $\mathcal{C}$.

Since

$$\mathbf{v} = P_{\mathcal{B}}[\mathbf{v}]_{\mathcal{B}}$$

and

$$[\mathbf{v}]_{\mathcal{C}} = P_{\mathcal{C}}^{-1}\mathbf{v},$$

we obtain

$$[\mathbf{v}]_{\mathcal{C}} = P_{\mathcal{C}}^{-1}P_{\mathcal{B}}[\mathbf{v}]_{\mathcal{B}}.$$

Therefore, the change-of-basis matrix is

$$P_{\mathcal{C} \leftarrow \mathcal{B}} = P_{\mathcal{C}}^{-1}P_{\mathcal{B}}.$$

Example: Changing Coordinates

Let

$$\mathcal{B} = \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ -1 \end{bmatrix} \right\}$$

and let $\mathcal{E}$ be the standard basis.

Then

$$P_{\mathcal{B}} = \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Suppose

$$[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 2 \\ 1 \end{bmatrix}.$$

Then

$$\mathbf{v} = P_{\mathcal{B}}[\mathbf{v}]_{\mathcal{B}} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Hence,

$$\boxed{[\mathbf{v}]_{\mathcal{E}} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}.$$

Linear Transformations in Different Bases

Suppose

$$T : V \rightarrow V$$

is a linear transformation. Its matrix depends on the chosen basis.

Let

$$A_{\mathcal{B}}$$

represent T in basis $\mathcal{B}$, and

$$A_{\mathcal{C}}$$

represent the same transformation in basis $\mathcal{C}$.

Then

$$A_{\mathcal{C}} = P_{\mathcal{C}}^{-1} A_{\mathcal{B}} P_{\mathcal{C}}.$$

Thus, changing the basis changes the matrix representation, not the underlying transformation.

Similar Matrices

Two square matrices A and B are **similar** if

$$B = P^{-1}AP$$

for some invertible matrix P .

Similar matrices represent the same linear transformation under different coordinate systems.

Important properties are preserved under similarity, including

$$\det(A) = \det(B).$$

The eigenvalues are also preserved.

Why Change the Basis?

A suitable basis can make a linear transformation easier to understand or compute. For example, a matrix may have a complicated representation in the standard basis but become diagonal in another basis.

This motivates the idea of

choosing coordinates that simplify the problem.

This idea leads directly to

eigenvectors and diagonalization.

Eigenvectors

Let

$$A : \mathbb{R}^n \rightarrow \mathbb{R}^n$$

be a linear transformation.

A non-zero vector $\mathbf{v}$ is an **eigenvector** of A if

$$A\mathbf{v} = \lambda\mathbf{v}$$

for some scalar λ .

The scalar λ is called the corresponding **eigenvalue**.

Thus, an eigenvector is a direction that is preserved by the transformation, although its magnitude and possibly its orientation may change.

Meaning of an Eigenvalue

If

$$A\mathbf{v} = \lambda\mathbf{v},$$

then:

- ▶ $\lambda > 1$ stretches $\mathbf{v}$.
- ▶ $0 < \lambda < 1$ shrinks $\mathbf{v}$.
- ▶ $\lambda = 1$ leaves $\mathbf{v}$ unchanged.
- ▶ $\lambda = 0$ maps $\mathbf{v}$ to $\mathbf{0}$.
- ▶ $\lambda < 0$ reverses the direction.

The important property is that

$\mathbf{v}$ remains on the same line after transformation.

Example: An Eigenvector

Consider

$$A = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix}$$

and

$$\mathbf{v} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}.$$

Then

$$A\mathbf{v} = \begin{bmatrix} 2 & 0 \\ 0 & 3 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 2 \\ 0 \end{bmatrix}.$$

Therefore,

$$A\mathbf{v} = 2\mathbf{v}.$$

Hence,

$\mathbf{v}$ is an eigenvector with eigenvalue $\lambda = 2$.

Finding Eigenvalues

Starting from

$$A\mathbf{v} = \lambda\mathbf{v},$$

we obtain

$$A\mathbf{v} - \lambda\mathbf{v} = \mathbf{0}.$$

Therefore,

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

For a non-zero solution $\mathbf{v}$ to exist,

$$\det(A - \lambda I) = 0.$$

This equation is called the

characteristic equation.

Its roots are the eigenvalues of A .

Characteristic Polynomial

The polynomial

$$p(\lambda) = \det(A - \lambda I)$$

is called the **characteristic polynomial**.

The eigenvalues satisfy

$$p(\lambda) = 0.$$

For an $n \times n$ matrix, the characteristic polynomial has degree n .

Therefore, finding eigenvalues involves:

Matrix $\rightarrow$ Characteristic polynomial $\rightarrow$ Roots

Example: Finding Eigenvalues

Consider

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}.$$

The characteristic equation is

$$\det(A - \lambda I) = 0.$$

Thus,

$$\det \begin{bmatrix} 2 - \lambda & 1 \\ 1 & 2 - \lambda \end{bmatrix} = 0.$$

Therefore,

$$(2 - \lambda)^2 - 1 = 0.$$

Hence,

$$(\lambda - 1)(\lambda - 3) = 0.$$

The eigenvalues are

$$\boxed{\lambda_1 = 1, \quad \lambda_2 = 3.}$$

Finding Eigenvectors

Once an eigenvalue λ is known, substitute it into

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

The non-zero solutions form the eigenvectors associated with λ .

For

$$A = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix},$$

consider $\lambda = 3$:

$$A - 3I = \begin{bmatrix} -1 & 1 \\ 1 & -1 \end{bmatrix}.$$

Solving

$$(A - 3I)\mathbf{v} = \mathbf{0}$$

gives

$$\mathbf{v} = t \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad t \neq 0.$$

Eigenspace

For an eigenvalue λ , the corresponding **eigenspace** is

$$E_\lambda = \text{Null}(A - \lambda I).$$

Thus, an eigenspace contains

the zero vector

and all eigenvectors associated with λ .

For example, if

$$E_\lambda = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\},$$

then every non-zero vector of the form

$$t \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

is an eigenvector corresponding to λ .

Eigenvectors and the Null Space

The eigenvector equation can be written as

$$(A - \lambda I)\mathbf{v} = \mathbf{0}.$$

Therefore,

$$\mathbf{v} \in \text{Null}(A - \lambda I).$$

Hence,

$$E_\lambda = \text{Null}(A - \lambda I).$$

This connects eigenvectors directly to the concepts of null space and homogeneous systems.

Eigenvectors with Distinct Eigenvalues

Suppose

$$A\mathbf{v}_1 = \lambda_1 \mathbf{v}_1$$

and

$$A\mathbf{v}_2 = \lambda_2 \mathbf{v}_2,$$

where

$$\lambda_1 \neq \lambda_2.$$

Then $\mathbf{v}_1$ and $\mathbf{v}_2$ are linearly independent.

More generally,

Eigenvectors corresponding to distinct eigenvalues are linearly independent.

This result is important for constructing an eigenvector basis.

Eigenbasis

Suppose an $n \times n$ matrix has n linearly independent eigenvectors:

$$\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n.$$

Then these vectors form a basis of $\mathbb{R}^n$.

Construct

$$P = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & & | \end{bmatrix}.$$

The eigenvectors provide a coordinate system in which the transformation becomes particularly simple.

Diagonalization

Let

$$P = \begin{bmatrix} | & | & \cdots & | \\ \mathbf{v}_1 & \mathbf{v}_2 & \cdots & \mathbf{v}_n \\ | & | & \cdots & | \end{bmatrix}$$

contain linearly independent eigenvectors.

Let

$$D = \begin{bmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n \end{bmatrix}.$$

Then

$$\boxed{A = PDP^{-1}.}$$

This is called **diagonalization**.

Why Diagonalization is Useful

For a diagonalizable matrix,

$$A = PDP^{-1}.$$

Therefore,

$$A^k = PD^kP^{-1}.$$

Since D is diagonal,

$$D^k = \begin{bmatrix} \lambda_1^k & 0 & \cdots & 0 \\ 0 & \lambda_2^k & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_n^k \end{bmatrix}.$$

Thus, powers of A can be computed through simple powers of its eigenvalues.

Geometric Interpretation

A general linear transformation may change both the direction and magnitude of a vector.

Eigenvectors identify special directions for which

$$A\mathbf{v} = \lambda\mathbf{v}.$$

The direction is preserved while the magnitude is scaled by $|\lambda|$.

Therefore, eigenvectors reveal the transformation's **intrinsic directions**.

Eigenvalue $\lambda = 0$

If

$$A\mathbf{v} = 0\mathbf{v} = \mathbf{0},$$

for a non-zero $\mathbf{v}$, then

$$\lambda = 0$$

is an eigenvalue.

This means that

$$\mathbf{v} \in \text{Null}(A).$$

Therefore,

$$0 \text{ is an eigenvalue} \iff \text{Null}(A) \neq \{\mathbf{0}\}.$$

For a square matrix, this is also equivalent to

$$\det(A) = 0.$$

Eigenvalues and Invertibility

For a square matrix A ,

A is invertible

if and only if

$$\det(A) \neq 0.$$

Since

$$\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n,$$

when the eigenvalues are counted with algebraic multiplicity,

$$A \text{ is invertible} \iff 0 \text{ is not an eigenvalue.}$$

Thus, determinants and eigenvalues provide two different ways to study invertibility.

Trace and Eigenvalues

The **trace** of a square matrix is the sum of its diagonal entries:

$$\operatorname{tr}(A) = a_{11} + a_{22} + \cdots + a_{nn}.$$

The trace is equal to the sum of the eigenvalues:

$$\operatorname{tr}(A) = \lambda_1 + \lambda_2 + \cdots + \lambda_n.$$

Similarly,

$$\det(A) = \lambda_1 \lambda_2 \cdots \lambda_n.$$

These relationships connect eigenvalues with two fundamental matrix quantities.

Eigenvectors of Symmetric Matrices

A real matrix is symmetric if

$$A = A^T.$$

For real symmetric matrices:

- ▶ all eigenvalues are real,
- ▶ eigenvectors corresponding to distinct eigenvalues are orthogonal,
- ▶ an orthonormal basis of eigenvectors exists.

Therefore, a real symmetric matrix can be written as

$$A = QDQ^T,$$

where Q is an orthogonal matrix.

Applications of Eigenvalues

Eigenvalues and eigenvectors appear throughout computing.

- ▶ **Principal Component Analysis (PCA)**
- ▶ **Dimensionality reduction**
- ▶ **Image processing**
- ▶ **Graph and network analysis**
- ▶ **Differential equations**
- ▶ **Stability analysis**

The central idea is to identify directions in which a transformation acts by simple scaling.

Linear Algebra in Computing

Linear algebra provides a mathematical language for representing and transforming structured data.

Common objects include

- ▶ vectors for features and observations,
- ▶ matrices for transformations and datasets,
- ▶ linear systems for solving computational problems,
- ▶ eigenvectors for discovering important directions.

These ideas form part of the mathematical foundation of

Computer Graphics, Machine Learning, AI and Data Science.

Vectors as Data Representation

A data instance can be represented as a vector.

For example, a student may be represented by

$$\mathbf{x} = \begin{bmatrix} \text{age} \\ \text{study hours} \\ \text{attendance} \\ \text{previous score} \end{bmatrix}.$$

A collection of observations can then be represented as a matrix

$$X = \begin{bmatrix} - & \mathbf{x}_1^T & - \\ - & \mathbf{x}_2^T & - \\ \vdots & \vdots & \\ - & \mathbf{x}_m^T & - \end{bmatrix}.$$

Data $\longrightarrow$ Vectors $\longrightarrow$ Matrices.

Feature Vectors in Machine Learning

In machine learning, an observation is often represented by a **feature vector**.

For example,

$$\mathbf{x} = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$$

may contain n features.

A dataset containing m observations can be represented by

$$\boxed{X \in \mathbb{R}^{m \times n} .}$$

Here,

m = number of observations

and

n = number of features.

This representation allows mathematical operations to be applied to entire datasets.

Linear Models

A simple linear model can be written as

$$y = \mathbf{w}^T \mathbf{x} + b.$$

Here,

$\mathbf{x}$ = input feature vector,

$\mathbf{w}$ = weight vector,

and

b = bias.

The dot product

$$\mathbf{w}^T \mathbf{x}$$

combines the input features using their corresponding weights.

Systems of Linear Equations

Many computational problems can be expressed as

$$Ax = b.$$

Here,

A = coefficient matrix, x = unknown vector, b = known vector.

Applications include

- ▶ solving networks of equations,
- ▶ resource allocation,
- ▶ circuit models,
- ▶ parameter estimation,
- ▶ numerical simulation.

Gaussian elimination provides a systematic method for solving such systems.

Linear Transformations in Computer Graphics

Vectors and matrices are fundamental to computer graphics. A point or direction can be represented as

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}.$$

A matrix can transform the vector:

$$\mathbf{x}' = A\mathbf{x}.$$

Different matrices can represent operations such as

- ▶ rotation,
- ▶ scaling,
- ▶ reflection,
- ▶ projection.

Thus, linear transformations provide a mathematical framework for manipulating geometric objects.

Rotation as a Matrix Transformation

A two-dimensional rotation through angle θ can be represented by

$$R = \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}.$$

For a vector $\mathbf{x}$,

$$\mathbf{x}' = R\mathbf{x}.$$

The transformation changes the direction of the vector while preserving its length.

$$\|\mathbf{x}'\| = \|\mathbf{x}\|.$$

Matrix transformations therefore provide a compact way to describe geometric operations.

Matrices and Images

A grayscale image can be represented as a matrix

$$I = \begin{bmatrix} p_{11} & p_{12} & \cdots & p_{1n} \\ p_{21} & p_{22} & \cdots & p_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ p_{m1} & p_{m2} & \cdots & p_{mn} \end{bmatrix},$$

where each entry represents a pixel intensity.

For colour images, multiple matrices can represent different colour channels.

Therefore,

Image data can be represented using matrices.

Linear algebra can then be used for image transformations, compression and analysis.

Principal Component Analysis

Principal Component Analysis (PCA) uses eigenvectors and eigenvalues to identify important directions in data.

Given a covariance matrix

$$C,$$

the principal directions are obtained from

$$C\mathbf{v} = \lambda\mathbf{v}.$$

The eigenvectors represent principal directions, while the corresponding eigenvalues indicate the amount of variation associated with those directions.

PCA can therefore be used for

dimensionality reduction.

PCA: Geometric Intuition

Suppose data points are distributed mainly along one direction.
Instead of describing the data using the original coordinates, PCA finds a new coordinate system aligned with the dominant directions.

Original coordinates $\longrightarrow$ Principal coordinates

This can reduce the number of dimensions while retaining important variation in the data.

The new directions are determined by eigenvectors of the covariance matrix.

Dimensionality Reduction

Suppose

$$\mathbf{x} \in \mathbb{R}^n$$

contains many features.

If the data can be represented approximately using $k < n$ important directions, we can project it into

$$\mathbb{R}^k.$$

Thus,

$$n \text{ dimensions} \longrightarrow k \text{ dimensions}, \quad k < n.$$

Benefits can include

- ▶ reduced computational cost,
- ▶ lower storage requirements,
- ▶ visualization of high-dimensional data,
- ▶ removal of less informative directions.

Linear Algebra in Neural Networks

A basic neural-network layer can be represented as

$$z = Wx + b.$$

Here,

W = weight matrix, x = input, b = bias.

A nonlinear activation function is then applied:

$$a = f(z).$$

Therefore, neural networks repeatedly combine

matrix multiplication + vector addition + nonlinear transformation.

Vectors and Embeddings

In modern AI systems, objects such as

- ▶ words,
- ▶ sentences,
- ▶ images,
- ▶ documents,

can be represented as vectors called **embeddings**.

For example,

$$\mathbf{x} \in \mathbb{R}^d.$$

Relationships between embeddings can be studied using vector operations such as

$$\mathbf{x} \cdot \mathbf{y}$$

and

$$\cos \theta = \frac{\mathbf{x} \cdot \mathbf{y}}{\|\mathbf{x}\| \|\mathbf{y}\|}.$$

These provide measures of directional similarity.

Matrices in Graphs and Networks

A graph can be represented using matrices.

For example, an adjacency matrix records connections between vertices:

$$A_{ij} = \begin{cases} 1, & \text{if vertices } i \text{ and } j \text{ are connected,} \\ 0, & \text{otherwise.} \end{cases}$$

Thus,

Graph $\longrightarrow$ Matrix representation.

Matrix operations and eigenvalue methods can then be used to study structural properties of networks.

Rank and Redundant Information

Suppose a dataset is represented by

$$X \in \mathbb{R}^{m \times n}.$$

If some columns are linear combinations of other columns, then the features are not all independent.

The rank measures the number of independent directions:

$$\text{rank}(X) = \text{number of independent directions.}$$

A rank smaller than the number of features indicates linear dependence or redundancy in the representation.

Linear Algebra and Numerical Computing

Many computational algorithms repeatedly perform operations such as

$$A\mathbf{x}, \quad A\mathbf{x} = \mathbf{b}, \quad A^{-1}, \quad \det(A).$$

In practice, efficient numerical algorithms are preferred over direct symbolic calculations. Examples include

- ▶ Gaussian elimination,
- ▶ LU decomposition,
- ▶ QR decomposition,
- ▶ iterative methods.

Thus, linear algebra connects mathematical models with efficient computational procedures.

Linear Algebra Across AI

Concept	Example Application
Vectors	Feature representation
Matrices	Datasets and transformations
Dot products	Similarity and linear models
Linear systems	Numerical modelling
Determinants	Invertibility and geometry
Eigenvectors	PCA and spectral methods
Change of basis	Coordinate transformations
Rank	Dimensionality and redundancy
Cross products	3D geometry

Linear algebra therefore provides a common mathematical foundation across many areas of computing and AI.

Conclusion

The central idea of linear algebra is the study of

vectors, spaces, and transformations.

Matrices provide a computational representation of linear transformations.

Systems, determinants, vector spaces, dot products, cross products, change of basis, and eigenvalues are different perspectives on these same underlying structures.

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Thank You

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